Empirical Study of Memorization in Differentially Privately Fine-Tuned Large Language Models

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Link to GitHub Repo

Abstract

Large language models (LLMs) trained on sensitive data risk unintentionally memorizing private information, posing significant privacy concerns. Differential privacy (DP) offers a rigorous mathematical framework to mitigate this risk, but its effectiveness in reducing memorization in fine-tuned LLMs remains underexplored. In this paper, we empirically investigate the relationship between the privacy budget ε and canary memorization in GPT-2 Small fine-tuned with DP-SGD on a mental health Reddit dataset. Using the canary insertion technique, we insert synthetic sensitive sequences at varying frequencies and measure memorization by the exposure metric across combinations of $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ and insertion frequencies $\mathrm{freq} \in \{1, 5, 10, 50\}$. Our results show that without DP, the model memorizes canaries with exposure scaling with insertion frequency, reaching the theoretical maximum of 19.78 bits. With any finite ε, memorization is effectively suppressed to near-random levels regardless of frequency. The utility cost of DP is a 39-44% increase in validation perplexity over the no-DP baseline. We conclude that DP must be applied, but the specific value of ε has limited impact on memorization within the tested range.

1. Introduction

In 2017, a group of researchers at Google Brain proudly announced the architecture of the transformer neural model, which revolutionized the field of Natural Language Processing (NLP) and led to the rise of large language models (LLMs) (Vaswani et al., 2017). Training LLMs requires large amounts of data, and if not curated carefully, these datasets might contain sensitive information such as social security numbers or medical records. This poses a severe problem in model security and ethics, as shown by Carlini et al. (2019) that neural networks have a high risk of unintentionally memorizing fine-grained information and oddly-specific details. To mitigate this issue and protect the privacy of individuals, one approach is to leverage methods from *differential privacy* (DP), which provides a rigorous mathematical framework for quantifying privacy (Dwork et al., 2006). With DP, researchers have developed DP-based training methods, such as DP-SGD, to train neural networks and have shown that they effectively mitigate privacy concerns during training. Background information about DP and DP-SGD utilized in this paper is described in Section 2.

Despite the prominent performance of DP-based training methods, the use of these methods in training or fine-tuning LLMs still remains underexplored in both research and industrial settings. Specifically, the benefits of DP in reducing unintentional memorization of training data are to be precisely quantified. Therefore, this study aims to narrow down this gap by investigating the following research question:

At what privacy budget ε does DP-SGD effectively prevent memorization of sensitive sequences in fine-tuned LLMs, and what is the accuracy cost of that protection?

2. Background

Differential privacy is a mathematical framework that leverages probability theories to rigorously quantify privacy in algorithms (Dwork et al., 2006). Formally, let $\mathcal{X}$ denote the data space and $X \in \mathcal{X}^*$ be a dataset. We say that two datasets X and X' are *neighboring*, denoted $X \sim X'$, if one can be obtained from the other by adding or removing a single record. With this setup, we say that a randomized algorithm $\mathcal{M}$ is (ε, δ)*-differentially private* if for all neighboring datasets $X \sim X'$, and all $S \subseteq \Omega$ where Ω is the output space of $\mathcal{M}$,

$$\mathbb{P}(\mathcal{M}(X) \in S) \leq e^{\varepsilon}\mathbb{P}(\mathcal{M}(X') \in S) + \delta.$$

In the context of training deep neural networks, we can adapt this definition and make gradient descent a differentially private algorithm so that the trained models do not memorize specific details of any training data. Defining $\mathcal{L}$ as the loss function, α as the learning rate, instead of updating the weights by $\alpha \nabla \mathcal{L}$ at every step, we first compute each per-sample gradient $\nabla \mathcal{L}_i$ and clip it to the L2 norm bound C, then sum the clipped per-sample gradients and add random noise from $\mathcal{N}(0, \sigma^2 C^2 \boldsymbol{I})$ to the sum before updating. By choosing the parameters σ and C strategically, one can show that this algorithm, named *differentially private stochastic gradient descent* (DP-SGD), satisfies (ε, δ)-differential privacy (Abadi et al., 2016).

3. Related Work

Since the introduction of the concept of DP, most previous work focused on using DP techniques on training or fine-tuning neural networks of simpler architectures compared to transformers. For example, on language modeling, McMahan et al. (2018) applied a DP algorithm called *federated averaging* (FedAvg) on training recurrent language models such as RNNs and LSTMs. In the study, they experimented with various hyperparameters for noise and gradient clipping to quantify the downside of DP training on the performance of language models. On the other hand, the study by Carlini et al. (2019) conducted experiments on investigating language modeling ability of LSTM models on the Penn Treebank dataset. A similar characteristic shared across these two papers is that the models they used in their experiments are rather small: 600,000 parameters for the model in Carlini et al. (2019) and 1.35M parameters for the model in McMahan et al. (2018). Since LLMs have grown exponentially in size nowadays, there is an urgent need for more research on deploying DP techniques on training larger models.

Studies on applying DP techniques on fine-tuning LLMs do exist, with Yu et al. (2024) being a foundational one. In this paper, the authors conducted two experiments, one on natural language understanding (NLU) and the other on natural language generation (NLG), both using DP-SGD, and they compared the model performance against the state-of-the-art on different benchmarks. The models they used come from the GPT-2 and RoBERTa suites, with GPT-2 Small being the smallest model but still having 117M parameters which is much larger than the models in previous studies. More recently, a paper by Du et al. (2025) also examined the effect of DP-SGD in fine-tuning GPT-2 models. However, they focused more on comparing different fine-tuning methods, including FFT, LoRA, prefix-tuning, and P-tuning, rather than directly investigating DP-SGD on GPT-2.

Despite these studies on DP fine-tuning LLMs, the datasets they used to evaluate their models are not inherently private: Yu et al. (2024) used data from the restaurant domain and DART which is open-domain, while Du et al. (2025) used Wikitext-2-v1 and AG News. The downside is that all these datasets are public sources and do not contain sensitive information. In order to understand how DP techniques perform on private datasets, which was the reason DP was developed in the first place, it is preferred to apply them to a more private, sensitive domain so that we can truly evaluate the effectiveness of DP before its actual deployment in real-world settings. This is thus the main motivation of this project: applying DP techniques to a mental health dataset, which is inherently privacy-sensitive.

4. Methodology

4.1. Model

In this project, we used the GPT-2 Small model developed by OpenAI (Radford et al., 2019). GPT-2 Small is a decoder transformer model that transforms an input directly to the output, unlike BERT which outputs embeddings of the input sequence. As such, GPT-2 is known for its ability in language generation, which matches the objective of this study. The GPT-2 Small model consists of an embedding layer with positional encoding, 12 transformer blocks, and an output layer, adding up to 117M parameters and thus is a great balance of performance and computational cost.

4.2. Data

The data used in this project is a Reddit Mental Health Dataset available for downloads on Hugging Face. There are 151,228 rows in total in the dataset, each row being a post related to mental health containing attributes such as the user, the text of the post, the time posted, and more. Only the text of the posts is of interest in this project. Other attributes are discarded. From the raw dataset, we randomly sampled 10,000 posts to form the fine-tuning set, and 1,000 more as the validation set.

4.3. Experimental Design

In the study by Carlini et al. (2019), the authors proposed the method of *canary insertion*. In this context, a *canary* refers to a piece of text purposefully inserted into the training data in order to measure the extent to which the model memorizes the training data instead of learning the underlying patterns. In their study, they used canaries of the form

> *The random number is …*

and fill in the blank with 9-digit numbers. By varying the number of times the canaries were inserted and the privacy budget ε in DP-SGD, the authors were able to quantify how much DP helps with reducing unintentional memorization of the training data.

In the current project, we adopted the same design and applied it to our mental health dataset with the following form of canaries:

> *My patient ID is …*

and fill in the blank with a random 6-digit number representing a patient ID. The reason for choosing this canary form is that it is directly related to mental health, which is the theme of the dataset, and is inherently private information that a patient would not wish to reveal which is the main motivation of the project.

With this canary form, we varied the number of times the canaries were inserted into the training set, with $\text{freq} = 1, 5, 10, 50$. We also pre-specified some values of privacy budget to conduct the experiment, with $\varepsilon = 0.5, 1, 2, 4, 8, \infty$, where $\varepsilon = \infty$ means vanilla gradient descent. For each pair of frequency and privacy budget ε, we fine-tuned GPT-2 Small using the training set with canaries inserted for 3 epochs. The Python library `Opacus` was used for DP-SGD support. Each model was then assessed by calculating the *log-perplexity* on the validation set as a proxy for model utility, mathematically expressed as

$$\text{log-perp}_\theta(x_1, \cdots, x_n) = -\frac{1}{n} \sum_{i=1}^{n} \log \mathbb{P}(x_i | f_\theta(x_1, \cdots, x_{i-1})).$$

To see more fine-grained differences between different privacy budget ε, we report the validation *perplexity* instead, expressed as

$$\text{perp}_\theta(x_1, \cdots, x_n) = \exp(\text{log-perp}_\theta(x_1, \cdots, x_n)).$$

To measure unintentional memorization of the training data, we also computed the *exposure* of each model as

$$\text{exposure} = \log_2(900000) - \log_2(\text{rank}),$$

where rank measures the rank of the probability that the random number in the canaries appears among all 900,000 possible 6-digit numbers. A small exposure means that the model does not memorize the canaries and the output distribution is generated entirely by chance, while $\text{rank} = 1$ and $\text{exposure} = \log_2(900000) \approx 19.78$ indicates that the model memorizes the canaries perfectly (Carlini et al., 2019). Supported by the definition of (ε, δ)-DP, a key consequence is that we have a formal upper bound on this metric. By definition (Dwork et al., 2006), (ε, δ)-DP bounds the log-likelihood ratio between any model output under neighboring datasets to at most ε (up to a δ failure probability), which directly implies that the expected canary exposure is bounded by approximately $\varepsilon / \log 2$ bits for small δ. In other words, a smaller privacy budget ε directly limits how much advantage an adversary gains from the presence of a canary in the training data.

It is worth noting that not all parameters in the GPT-2 Small model were fine-tuned. In the project, we froze the embedding & positional encoding layer as well as the transformer head, and only fine-tuned the 12 transformer blocks. This was necessary due to GPU memory constraints, consistent with the approach of Yu et al. (2024).

5. Results & Discussion

5.1. Main Results

Figure 1 shows the line plot of validation perplexity vs privacy budget $\varepsilon = 0.5, 1, 2, 4, 8, \infty$ used in this project. From the plot, we see that for all privacy budgets ε, the effect from the frequency of canaries inserted into the training set is negligible, as the validation perplexities are mostly indistinguishable across frequencies. This is expected because the

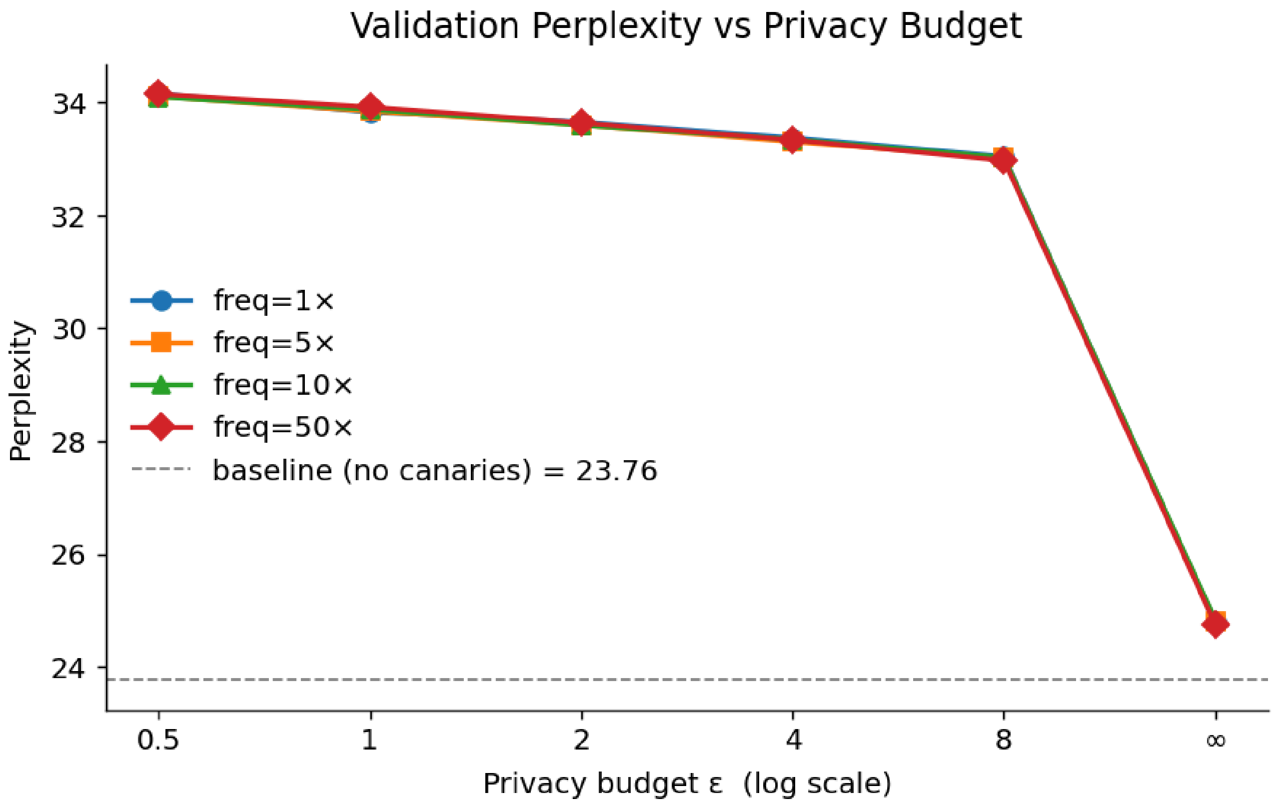

Figure 1: Validation perplexity vs privacy budget ε.

frequencies of canaries do not affect much how the model understands language during fine-tuning. Furthermore, we notice that the higher the privacy budget ε, the smaller the validation perplexity, which is also expected as a higher privacy budget ε means less privacy and thus the model can learn more patterns from the training data. These results are further compared against those from Yu et al. (2024) in Table 1.

In Table 1, note that the validation perplexities across the two studies are not comparable, as this metric is highly dependent on the validation data. However, as a general trend, we see that DP models perform worse than non-DP models in both studies, corroborating the plot shown in Figure 1. One discrepancy between the studies is that Yu et al. (2024) obtained lower percent increase from baselines, averaging to 26.1% while ours are 40.3% and 39.0% for comparable privacy budgets $\varepsilon = 4, 8$ in this study. This difference can be explained by various factors, the most prominent being that their training dataset consists of 42K samples, which is larger than ours and so the models might have learned to generalize better even with DP.

Figure 2 shows the line plot of canary exposure vs privacy budget ε. It is clear that unintentional memorization of the canaries occurs at $\varepsilon = \infty$, or no DP, even when the canary is inserted only once. The more times the canaries are inserted, the higher the canary exposure, with a maximum possible canary exposure of 19.78 achieved at freq $= 10$ and freq $= 50$. This result highlights a significant privacy risk, as if sensitive personal information is leaked into the training dataset, then the model is shown to more or less memorize that information if trained without privacy, which poses a significant threat to individual privacy. On the other hand, we notice that DP significantly reduces privacy concerns. With $\varepsilon = 8$, the model does not appear to memorize the canaries even when the canaries are inserted 50 times, since the canary exposure does not increase as the privacy budget ε increases from 0.5 to 8 for all frequencies. This is consistent with the

Model	Validation Perplexity	% Increase from Baseline
Yu et al. (2024), $\varepsilon = 6$		
GPT-2 Small	Not reported	-
GPT-2 Small + DP	4.51	N/A
GPT-2 Medium	3.19	-
GPT-2 Medium + DP	4.02	26.0%
GPT-2 Large	3.06	-
GPT-2 Large + DP	3.87	26.5%
GPT-2 XL	3.01	-
GPT-2 XL + DP	3.79	25.9%
This Project		
Baseline (no canaries, $\varepsilon = \infty$)	23.76	-
Baseline (with canaries, $\varepsilon = \infty$)	24.80	-
$\varepsilon = 0.5$	34.13	43.7%
$\varepsilon = 1.0$	33.87	42.6%
$\varepsilon = 2.0$	33.63	41.5%
$\varepsilon = 4.0$	33.34	40.3%
$\varepsilon = 8.0$	33.02	39.0%

Table 1: Validation perplexity with and without DP fine-tuning. For Yu et al. (2024), % increase is relative to the non-DP baseline of the same model size. For this project, % increase is relative to the no-canary baseline (23.76).

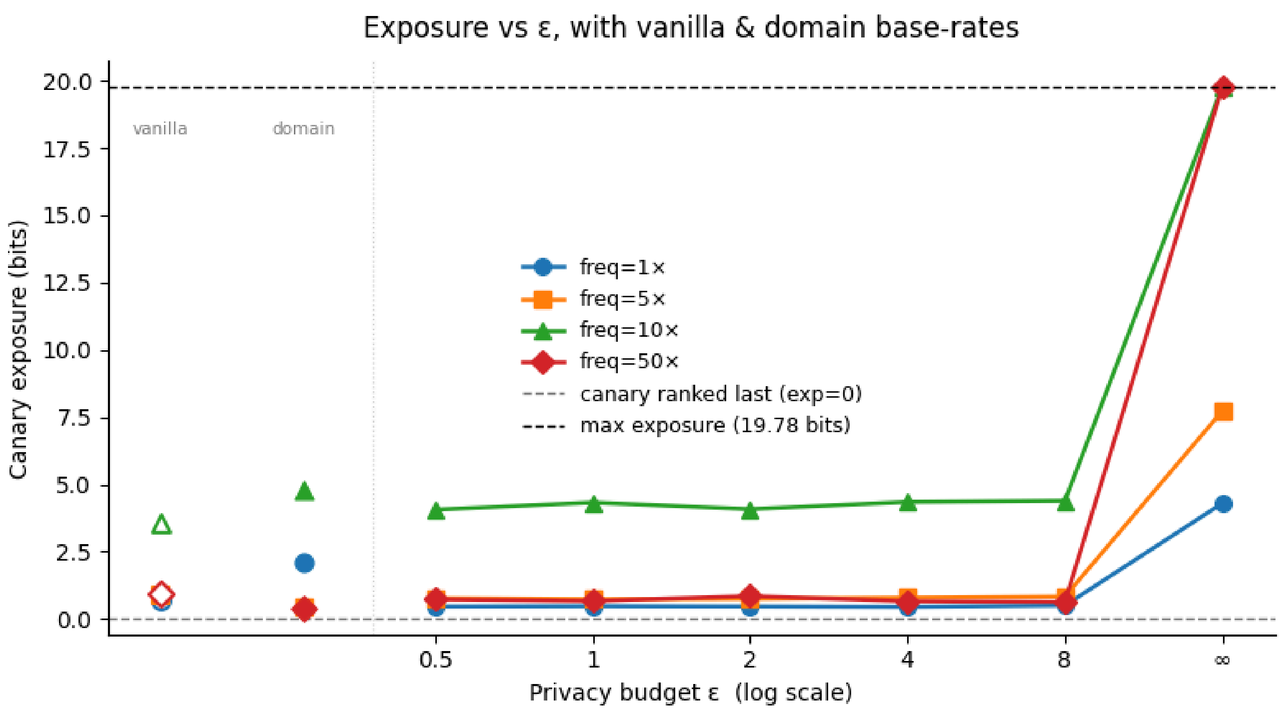

Figure 2: Canary exposure vs. privacy budget ε. Vanilla and domain base rates (pretrained only, and fine-tuned without canaries) are shown on the left.

theoretical bound of $\varepsilon / \log 2$ bits derived from the (ε, δ)-DP definition, as all observed exposures fall well below this limit. Altogether, we conclude that to reduce unintentional memorization, the privacy budget ε can be large, but it must exist.

5.2. Ablation Study

One noteworthy characteristic shown in Figure 2 is that at $\text{freq} = 10$ the canary exposure seems to start high compared to other frequencies. Therefore, we conducted an ablation study to see whether this is entirely by chance or a model artifact. Since we generated 4 canary numbers, one for each frequency, for each canary we also used the vanilla, not fine-tuned GPT-2 Small model and the domain fine-tuned GPT-2 without canaries and measured the canary exposure on these models. The results are shown in the left portion of Figure 2. Interestingly, the canary number 116632 for $\text{freq} = 10$ also has high canary exposure on the vanilla and domain GPT-2 Small models, even though these models have not seen this number in our experimental setup. Therefore, this phenomenon must be a model artifact from the original training of the GPT-2 Small model and thus does not affect our conclusion that DP effectively reduces unintentional memorization of the training data.

6. Conclusion

6.1. Summary

To summarize, this paper aimed to evaluate the effectiveness of using differential privacy techniques to train LLMs in reducing unintentional memorization of the training data. We employed the method of canary insertion proposed by Carlini et al. (2019), experimented with different combinations of canary insertion frequencies and privacy budgets ε, and evaluated the fine-tuned GPT-2 Small model using perplexity for model utility and canary exposure for privacy. Our results showed that the higher the privacy budget ε, the lower the validation perplexity, which is expected because the model can generalize better with less privacy. For canary exposure, we saw that the specific value of ε did not significantly affect the extent to which the canaries are memorized, nor did frequency if DP is used. We thus came to the conclusion that DP effectively reduces unintentional memorization in a mental health context, which directly answers our research question.

6.2. Limitations & Future Directions

There are several limitations associated with this study. First of all, we only experimented with the GPT-2 Small model due to limited computational resources, but this model is a relatively small and old LLM. More recent LLMs such as GPT-4 are much larger in size and have more parameters. Therefore, if resources permit, future researchers should study these larger LLMs in order to fully understand the benefits of DP in training LLMs. The dataset used in this project is also relatively small. DP training is computationally expensive, but future studies should attempt to include a larger dataset for both training and validation if possible so that we would be more confident in measuring model performance and generalizability.

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